Modeling Cross-Trial Context in Auditory Prediction with Recurrent Neural Networks

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Abstract

Auditory perception is commonly framed as a predictive process in which incoming sound is interpreted relative to internally generated expectations. Previous work has shown that auditory responses can reflect abstract sequence expectations beyond local stimulus statistics, but these accounts have focused primarily on prediction within a trial or short sequence. Here, we asked whether auditory prediction also depends on cross-trial context, and whether this contribution can be captured by recurrent neural networks. To test this, we trained participant-specific gated recurrent unit networks on an auditory deviance-detection task, allowing hidden states to carry over across trials, and evaluated them against a simplified predictive-coding prior baseline. We found that recurrent networks explained additional trial-level reaction-time variance beyond the prior baseline across participants, indicating that a static prior account was insufficient to capture the full structure of human behavior in this task. Notably, the most human-like models did not coincide with the most highly optimized models: behavioral alignment peaked before training loss reached its minimum. Cross-validated parameter maps further showed that successful fits occupied a shared region of model space rather than arising from arbitrary hyperparameter choices. Together, these results suggest that auditory prediction depends on temporally extended context carried across trial boundaries, and that recurrent neural networks provide a useful behavioral account of this contribution.

1 Introduction

Auditory perception is not a passive readout of sound events, but an inferential process in which incoming input is interpreted in light of prior context and experience (Friston, 2005; Friston & Kiebel, 2009). In predictive-coding accounts, perception depends on selecting the internal hypothesis that best predicts current sensory input, thereby minimising prediction error (Walsh et al., 2020).

In auditory perception, these ideas are often studied with sequence paradigms in which regular sound patterns are occasionally violated (Garrido et al., 2009; Winkler & Czigler, 2012). A repeated tone sequence can generate an expectation about the next sound, and a deviant tone can then be treated as evidence that the current hypothesis should be updated. By explicitly manipulating abstract sequence rules, Tabas et al. (2020) showed with 7T fMRI that responses in inferior colliculus, medial geniculate body and auditory cortex are shaped by abstract expectations rather than by local stimulus statistics alone. In that paradigm, participants knew that every sequence contained a deviant and that the deviant could occur only at positions 4, 5 or 6. Under these rules, the conditional probability of a deviant changed across the sequence, from $1/3$ at position 4 to $1/2$ at position 5 and 1 at position 6, allowing expectation to vary across positions despite the repeated standards. It is therefore possible to dissociate expectation-related responses from simple repetition effects, but this remains a within-trial account of auditory prediction.

An unresolved question is whether auditory prediction also depends on cross-trial context, that is, information carried forward from preceding trials into the processing of the current one (e.g., whether a position-4 versus position-6 deviant in one trial changes expectation in the next). Trial-wise modelling work has argued that prediction should be studied at the level of single-trial dynamics rather than condition averages alone (Lieder et al., 2013; Wacongne et al., 2012). Sequence-learning studies likewise suggest that auditory responses integrate regularities over multiple timescales, including recent observation history beyond the current local event structure (Maheu et al., 2019). This raises the possibility that what happens early in a block may shape how later sequences are processed, even when the within-trial stimulus structure is unchanged.

Recurrent neural networks provide a natural way to model this possibility because they process input sequentially while maintaining an internal state that is updated over time. In such models, current responses depend not only on the present stimulus, but also on information retained from earlier inputs. Recurrent architectures have therefore been used as models of temporally structured sensory processing, particularly for visual or more general sensory stimulation (Ali et al., 2022; Güçlü & van Gerven, 2017). In the present context, recurrence provides a way to formalise how latent context may persist across trial boundaries and influence the processing of the current sequence.

The present study asks whether cross-trial context contributes meaningfully to auditory prediction, and whether a recurrent state model can be used to capture that contribution in behavior and brain activity. More specifically, we ask:

RQ. Can participant-specific RNNs explain human trial-level reaction-time variability better than a simplified predictive-coding prior baseline?

For this question, we hypothesized that participant-specific RNNs would explain additional trial-level reaction-time variance beyond the simplified predictive-coding prior baseline.

2 Methods

2.1 Data

The study used data from 19 participants reported in Tabas et al. (2020). They were German native speakers (12 female), aged 24 to 34 years (mean 26.6), with no reported psychiatric, neurological, or hearing disorders.

Participants performed an auditory deviance-detection task based on short tone sequences. Each sequence contained seven repetitions of a standard tone and one deviant tone, and the deviant could occur only at positions 4, 5, or 6. The three tone frequencies were 1455, 1500, and 1600 Hz, which yielded six standard–deviant pairings across trials.

On each trial, participants listened to one sequence and reported the position of the deviant as quickly and accurately as possible using a three-button response box. This design made the critical task variable the evolving expectation over possible deviant positions within each sequence.

2.2 Model

Architecture We trained a family of gated recurrent unit (GRU) models to perform the same auditory deviance-detection task as the human participants. Each model received the tone sequence as a time-resolved input and was trained to classify the deviant position as 4, 5, or 6. The recurrent backbone was a one-layer GRU with a 512-dimensional hidden state, followed by a linear three-class readout. Different models corresponded to different combinations of sensory-encoding parameters, rather than to a single fixed architecture.

Input representation The auditory sequence was discretised into 10 ms time tokens. Thus, each 50 ms tone was represented by five consecutive tokens, and each 700 ms inter-stimulus interval by 70 tokens. Tone frequency was represented using an ERB-spaced Gaussian receptive-field (RF) population code. Each physical frequency f in Hz was first transformed to ERB rate,

$$\mathrm{ERB}(f) = 21.4 \log_{10}(1 + 0.00437 f).$$

The frequency range from 1300 to 1700 Hz was then divided into 128 equally spaced bins in ERB space. For example, a 1600 Hz tone corresponds to $\mathrm{ERB}(f) \approx 19.32$, which falls around the 99th bin of the 128-bin representation. This bin index c, rather than the raw frequency value, defined the centre of the input population code:

$$x_i = \exp\left[-\frac{1}{2}\left(\frac{i-c}{\sigma_{\mathrm{RF}}}\right)^2\right],$$

where i indexes ERB-spaced frequency channels and σ_{RF} controls the receptive-field width in bin units. Smaller σ_{RF} values produced sharper frequency-specific representations, whereas larger values produced broader tuning with greater overlap between neighbouring channels.

Sensory uncertainty was modelled by adding Gaussian noise directly to this encoded population vector. For each tone token, the noisy input was

$$\tilde{x}_i = \mathrm{clip}(x_i + \epsilon_i, 0, 1), \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2_{\mathrm{noise}}),$$

where σ_{noise} controls the standard deviation of channel-wise input noise. The noisy vector was clipped to the valid activation range $[0, 1]$. Thus, σ_{RF} controlled the deterministic tuning width, whereas σ_{noise} controlled token-level variability in the encoded sensory input.

Because recurrent states were propagated across trial boundaries, we additionally appended explicit start-of-trial and end-of-trial boundary signals to the input representation. These were implemented as two dedicated extra input channels, each expressed as a separate one-token event: a start-of-trial token carried value 1 on the start boundary channel and 0 elsewhere, and an end-of-trial token carried value 1 on the end boundary channel and 0 elsewhere. These boundary tokens were appended as distinct 10 ms tokens rather than being mixed into tone tokens. Their role was to provide an explicit segmentation cue to the network, so that trial boundaries did not have to be inferred solely from silence or latent state dynamics while cross-trial context was still allowed to persist.

Generative model for training stimuli Training sequences were generated from the same abstract task structure as the human experiment. Each trial contained eight tones, with seven repetitions of a standard frequency and one deviant tone. The deviant could occur only at position 4, 5, or 6, and the model target was the deviant position. Training examples were organised into blocks of 10 trials. Within each block, a single standard–deviant frequency pair was used across all 10 trials, matching the blocked structure of the human experiment. Deviant positions were not forced to be balanced within a block. Instead, balance was imposed across blocks, so that position-4, position-5 and position-6 deviants were equated over longer spans of training data while preserving the intended block-level structure. This prevented the model from solving the task by exploiting within-block position statistics and instead encouraged learning of the abstract sequential rule implemented in the experiment.

The three frequencies used in the human fMRI experiment were excluded from this generative process. This allowed the RNN to learn the task rule from a broad distribution of synthetic tone sequences rather than memorising the exact stimulus frequencies later used for human-model comparison.

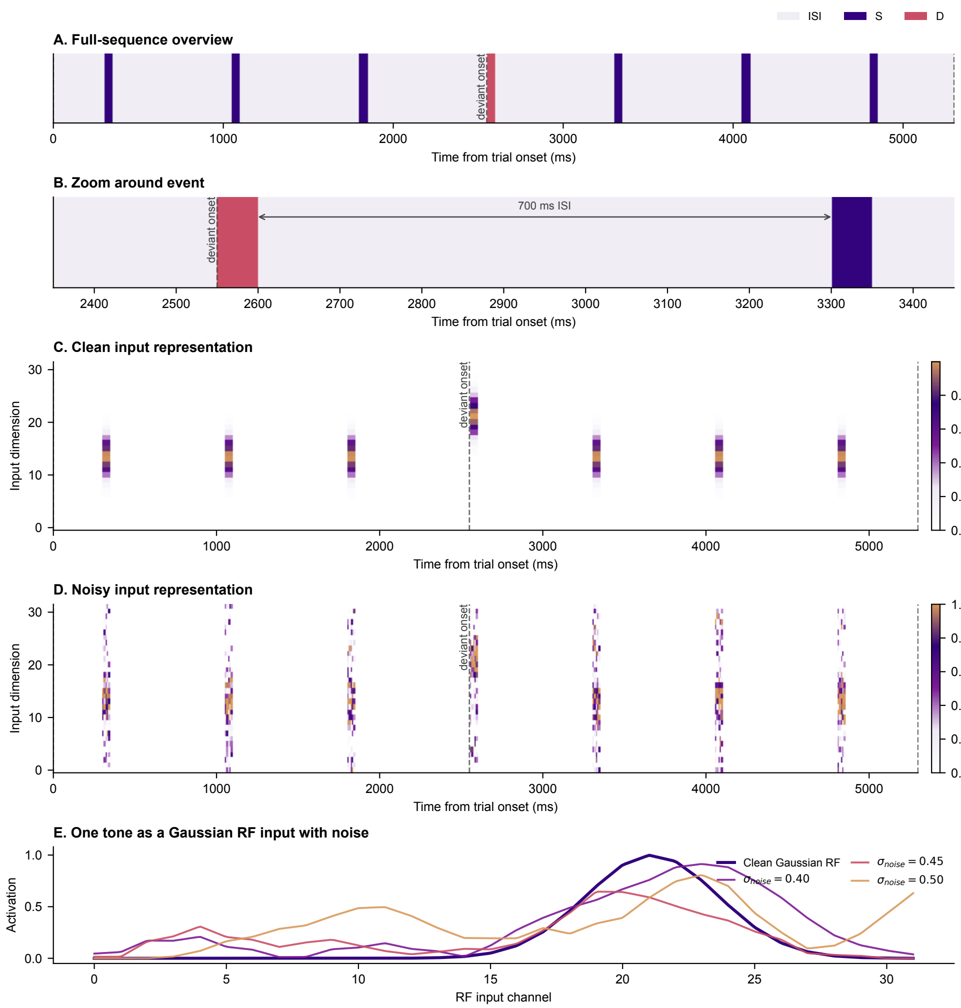

Figure 1: Schematic of the RNN input sequence. **A**, Categorical structure of a single trial, showing seven standards followed by one deviant. **B**, Zoomed view around deviant onset in the tokenised time series. **C**, Clean ERB-spaced receptive-field representation of the full trial. **D**, The same trial after additive Gaussian input noise. **E**, Effect of noise on one tone over its true 50 ms duration. Each token corresponded to 10 ms, so a 5300 ms trial contained approximately 530 recurrent time steps, plus separate boundary tokens when start-of-trial and end-of-trial markers were included. Silent inter-stimulus intervals were represented by empty input vectors, whereas tone periods contained frequency-tuned RF activations. The figure is illustrative and is intended to clarify the temporal format of the RNN input.

Training objective The model was trained with supervised cross-entropy to predict the deviant position from the evolving tone sequence. At each time token, the RNN produced three logits corresponding to deviant positions 4, 5, and 6. Training used two equally weighted supervision terms: a token-level loss over the supervised response window and a final-token loss at the end of the trial:

$$\mathcal{L} = -\frac{1}{|W|} \sum_{t \in W} \log p_\theta(y \mid h_t) - \log p_\theta(y \mid h_T),$$

where h_t is the GRU hidden state at token t, $y \in \{4, 5, 6\}$ is the true deviant position, W denotes the supervised response window, and T denotes the final token of the trial. The supervised response window began at the actual deviant onset and extended to the onset of the next tone. Given a 50 ms tone followed by a 700 ms interval, this window spanned 750 ms, corresponding to 75 tokens at the 10 ms sampling resolution.

Training used full backpropagation through time over continuous trial streams rather than truncated updates. Within each block, the hidden state was carried from one trial to the next instead of being reset at every trial (Zhang et al., 2020). Between blocks, the hidden state was re-initialized. Before entering the next trial, the carried state was multiplied by a fixed coefficient of 0.25. This damping step reduced how strongly previous-trial information was preserved while still allowing recent context to persist (Chien et al., 2021). Full BPTT was used because the loss on the current trial could depend on hidden-state information inherited from earlier trials; unrolling the full block therefore allowed those cross-trial dependencies to influence learning. Models were optimised with AdamW, learning rate 1×10^{-4}, and batch size 32. Training was stopped early if R^2_{model} failed to improve for five consecutive epochs, and the checkpoint with the highest observed R^2_{model} was retained. During evaluation, network weights were frozen and only the forward recurrent dynamics were retained.

To test how the sensory input parameterisation affected behavior, models were trained across a grid of Gaussian RF widths and input-noise levels. The RF-width sweep included $\sigma_{\text{RF}} \in \{10, 15, 20, 25, 30, 35, 40, 45, 50\}$. The input-noise sweep included $\sigma_{\text{noise}} \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$.

Behavioral readout and model selection For each trained checkpoint, we derived model reaction times from the time-resolved class probabilities. At each token after deviant onset, the softmax output assigned a probability $p_{\text{correct},t}$ to the correct deviant position. A model response was defined as the first token at which $p_{\text{correct},t}$ crossed a criterion threshold, indicating that the model was sufficiently confident to respond. This readout treats the response threshold as a model analogue of a behavioral decision criterion: higher thresholds correspond to more conservative responding, whereas lower thresholds correspond to less conservative responding (Li et al., 2024; Ratcliff et al., 2016). Thresholds were swept from 0.5 to 1.0.

Behavioral model selection was carried out hierarchically across checkpoints and RF/noise combinations. Checkpoint evaluation began only after validation accuracy exceeded 0.8. For every eligible checkpoint and threshold, we computed the trial-level Pearson correlation between model reaction time and human reaction time, denoted r_{model}, and converted it to

$$R^2_{\text{model}} = r^2_{\text{model}}.$$

We compared this against a simplified optimal-prior baseline, in which prior certainty was assigned according to deviant position: $P4 = 1/3$, $P5 = 1/2$, and $P6 = 1$. The baseline explained variance was

$$R^2_{\text{prior}} = r^2(RT_{\text{human}}, P_{\text{prior}}).$$

The main behavioral selection score was

$$\Delta R^2 = R^2_{\text{model}} - R^2_{\text{prior}}.$$

Candidate models were required to have $r_{\text{model}} > 0$ and $\Delta R^2 > 0$. Selection was then performed in a leave-one-participant-out manner. For a given held-out participant, the remaining participants ($n = 18$) served as validation data. Within each RF/noise combination, each eligible checkpoint was evaluated across 10 stochastic noise realizations under the same parameter setting, yielding a distribution of validation scores. The checkpoint-level score for that combination was defined as the mean validation ΔR^2 across these noise realizations and participants. The checkpoint with the highest validation score was selected as the representative candidate for that RF/noise combination. These per-combination winners were then compared across RF/noise combinations, and the combination with the highest validation score was selected for the held-out participant. This procedure yielded one participant-specific GRU model for subsequent analyses.

3 Results

3.1 More training made the model less human-like

Behavioral alignment and task optimisation peaked at different stages of training (Fig. 2). Mean behavioral ΔR^2 reached its maximum earlier, around optimizer step 875, whereas mean training loss continued to improve until around step 940. Beyond the behavioral optimum, additional training reduced loss but also reduced correspondence with human trial-level reaction times.

This divergence indicates that solving the task more accurately did not make the model more human-like. Instead, the checkpoint that best matched human behavior occupied an intermediate stage of training rather than the final loss minimum. Behavioral model selection therefore could not be replaced by standard training-loss selection, because the optimisation trajectory itself showed that lower loss and stronger human alignment came apart.

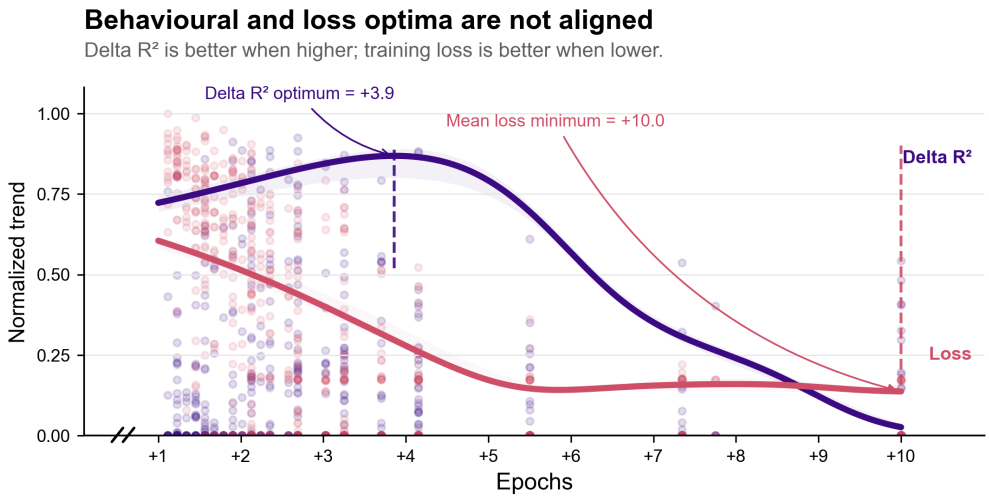

Figure 2: Behavioral alignment and training loss were not optimised at the same checkpoint. Mean validation ΔR^2 peaked earlier than mean training-loss improvement, indicating that continued task optimisation moved the model away from human-like trial-level timing.

3.2 Cross-validated participant-level fits revealed recurring parameter regimes

Cross-validated behavioral gains were not distributed randomly across the RF/noise space (Fig. 3). Instead, positive validation ΔR^2 values recurred in a limited set of RF/noise regions across participants, whereas other regions were consistently weak or negative. In particular, the highest-noise settings were generally unfavourable, while positive solutions clustered in a small number of intermediate regimes.

At the same time, the heatmaps were not identical across participants. Some participants showed stronger gains in lower-noise, broader-RF settings, whereas others favoured more intermediate combinations. The overall pattern was therefore structured but not uniform: the behavioral sweep converged on recurring solution families rather than arbitrary isolated cells, yet still preserved meaningful participant-level variation. This justified selecting models individually, while also suggesting that the behavioral gains reflected stable properties of the modelling regime rather than idiosyncratic overfitting.

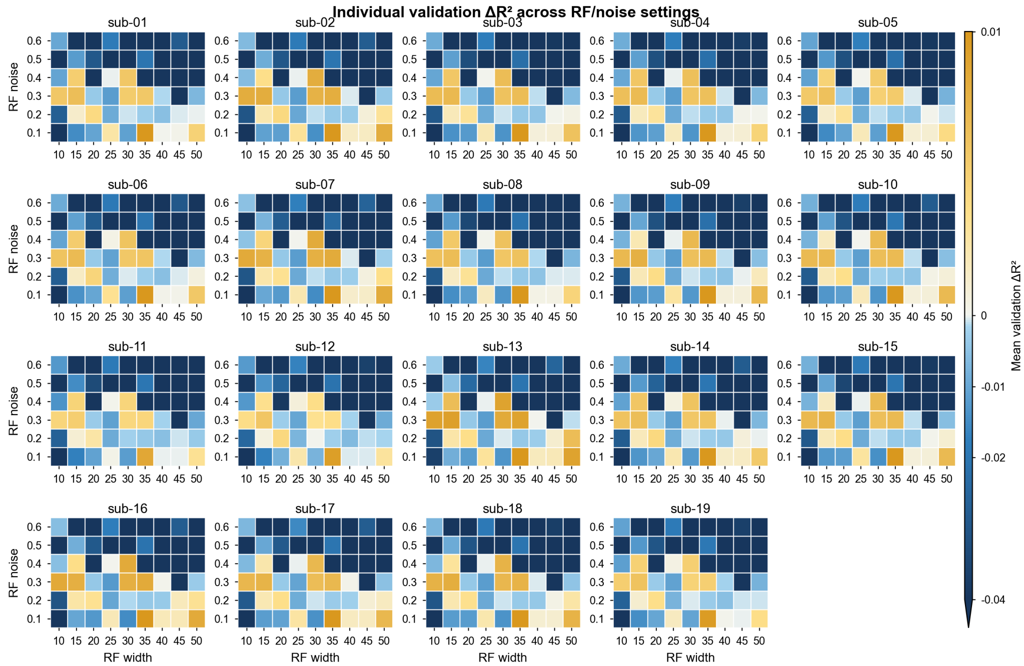

Figure 3: Participant-level cross-validated behavioral gains across RF/noise combinations. Positive validation ΔR^2 values recurred in a limited set of parameter regimes across participants, indicating structured but non-identical behavioral preferences.

3.3 Behaviorally selected recurrent models explained additional reaction-time variance beyond the prior baseline

We next asked whether the selected recurrent models explained more human reaction-time variance than the simplified prior baseline (RQ1). Candidate checkpoints were evaluated on each participant's formal-trial sequence, and the main selection criterion was incremental explained variance, $\Delta R^2 = a_{R^2} - c_{R^2}$, with $a_r > 0$ required so that positive explained variance reflected the correct direction of association.

This procedure identified a best valid participant-specific model for every participant (Fig. 4). Across participants, the mean model–human correlation was $a_r = 0.498$ (median $= 0.437$, range $= 0.303$–0.858). The selected recurrent models explained a mean of $a_{R^2} = 0.269$ of human RT variance (median $= 0.191$, range $= 0.092$–0.736). The prior baseline was also strong, with mean $c_{R^2} = 0.230$ (median $= 0.144$), but the recurrent model exceeded it for all participants.

The incremental gain was modest but consistent. Across participants, the mean additional explained variance was $\Delta R^2 = 0.0385$ (median $= 0.0338$, range $= 0.0070$–0.1135). Importantly, these gains were not restricted to participants with weak prior fits. Some of the largest improvements appeared in participants for whom the prior baseline was already strong, whereas others showed only small gains despite high overall R^2. Thus, the recurrent model did not merely rescue poor baseline cases. Rather, it captured a small but reliable component of trial-level behavioral variance that was not explained by the simplified predictive-coding prior alone.

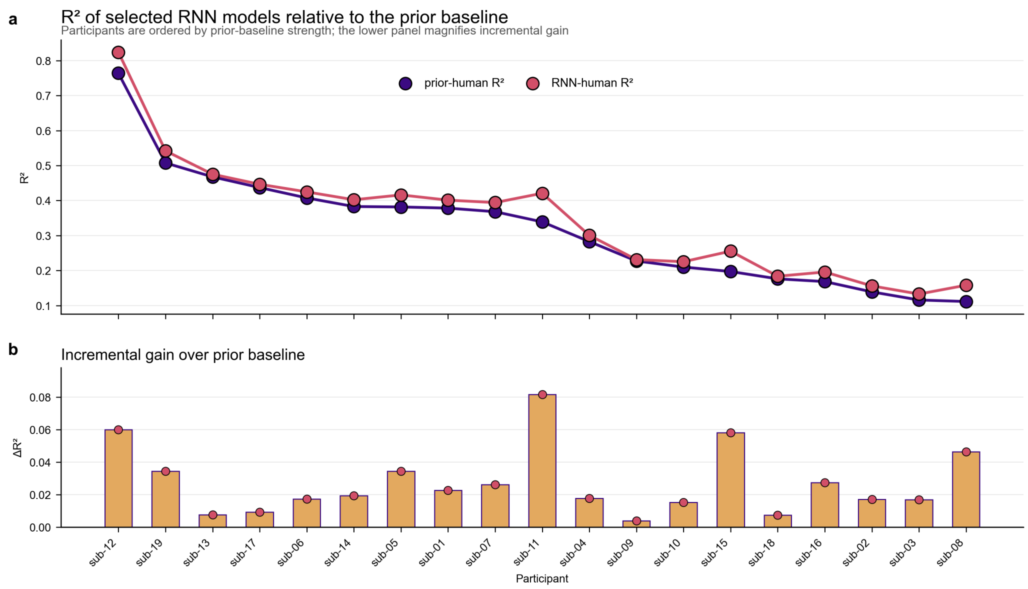

Figure 4: Participant-specific recurrent models explained more reaction-time variance than the prior baseline. The upper panel shows participant-wise R^2 for the selected recurrent model and the prior baseline; the lower panel shows the incremental gain, ΔR^2, over the prior model.

4 Discussion

The present results show that participant-specific recurrent models explain trial-level human reaction-time variability better than a simplified predictive-coding prior baseline. Importantly, the recurrent models did not replace the prior account. Rather, they explained additional behavioral variance beyond a baseline that already captured the main within-trial expectation structure of the task. The behavioral advantage therefore should not be read as evidence against predictive-coding-style expectation in this paradigm. Instead, it suggests that a static prior account is incomplete at the level of single-trial response timing.

The importance of this gain lies in what the prior baseline already contains. Because the prior model formalises the abstract within-trial expectation structure, any additional explanatory power from the recurrent model must arise from something not exhausted by that structure alone. The most plausible interpretation is that human behavior depends, at least in part, on temporally extended state information that persists across trials. The present analyses do not identify the exact form of that carried-over quantity, but they do show that allowing state dependence across trial boundaries improves behavioral explanation. This makes cross-trial context a concrete explanatory target rather than only a theoretical possibility. The interpretation is consistent with trial-wise approaches to auditory prediction (Lieder et al., 2013; Wacongne et al., 2012) and with evidence that human sequence learning integrates regularities over multiple timescales (Maheu et al., 2019).

A second implication concerns model selection. The most human-like checkpoint was not the most highly optimised checkpoint. Behavioral alignment peaked before training loss reached its minimum, which indicates that solving the task more efficiently was not equivalent to reproducing human response dynamics more faithfully. One interpretation is that continued optimisation pushed the model toward increasingly effective task solutions that no longer respected the constraints shaping human behavior. Under this view, the divergence between lower loss and higher behavioral fidelity is informative rather than problematic: it implies that human-like prediction in this task is associated with bounded or partially suboptimal internal dynamics, rather than with the globally best-performing solution. This interpretation resonates with recent arguments that more capable sequence models can become less human-like because they become too strong as predictors (Oh & Linzen, 2026). More broadly, it sharpens the logic of task optimisation in sensory modelling: optimisation may help uncover useful model structure, but the final optimum need not be the most behaviorally faithful point on the training trajectory (Ali et al., 2022; Kell et al., 2018).

The RF/noise heatmaps further qualify this behavioral correspondence. The participant-level maps were broadly similar rather than qualitatively different, with positive ΔR^2 values recurring in comparable regions of parameter space across participants. This suggests that the recurrent advantage was not driven by arbitrary hyperparameter choices. At the same time, the exact best setting was not identical for everyone. A reasonable interpretation is therefore that participant-specific selection occurred within a shared solution landscape: the common task structure and model family constrained models toward a limited set of viable behavioral regimes, while participant-level fitting identified the best local optimum within that shared region. This reading is consistent with the broader point that conclusions should not rest on a single trained network instance, because different instances can achieve similar external performance while differing internally (Mehrer et al., 2020). It is also consistent with the idea that complex models can contain sloppy parameter directions, such that exact best settings need not carry strong mechanistic meaning even when a stable effective region is present (Transtrum et al., 2015). The heatmaps therefore support a constrained landscape interpretation: the recurrent advantage was structured and repeatable across participants, but small local differences in best RF/noise settings should not be overinterpreted as one-to-one differences in underlying cognitive mechanism.

The current boundary of the evidence is that it is behavioral. These analyses establish that recurrent state trajectories are not a trivial re-description of the prior baseline, but they do not yet show whether those trajectories are neurally expressed or which latent variable best explains any additional neural variance. These questions will need to be addressed in future work. The present results should therefore be read as establishing the behavioral precondition for that next step. Because recurrent state dynamics improved trial-level behavioral explanation, it becomes justified to ask whether the same recurrent states are reflected in brain activity and whether their advantage is best interpreted in terms of dynamic belief-state computations.

A further limitation concerns biological plausibility. The models were trained with full backpropagation through time, which was useful here because the loss on the current trial could depend on state information inherited from earlier trials. However, this training procedure should be understood as an optimisation tool rather than as a literal account of how auditory systems learn across experience. The present claim is therefore narrower: recurrence provided a useful computational model of cross-trial dependence in behavior, not a biologically faithful learning rule. Future work could replace full BPTT with more biologically plausible recurrent-learning approximations. For example, Kalman-filter-based recurrent learning has been proposed as an alternative that can approach BPTT-like performance in sequence-learning benchmarks (Ali et al., 2022). Such approaches would be particularly relevant here because the model must update internal state across trial boundaries while learning from delayed behavioral targets.

A second limitation concerns the reaction-time readout. In the present analyses, model reaction times were derived using a fixed confidence threshold applied to the time-resolved probability of the correct response. This made the readout simple and interpretable, but it assumes a stable decision criterion across trials. Human reaction times are likely to reflect additional trial-by-trial variability in threshold, urgency, motor preparation and uncertainty. Future work could therefore explore richer decision readouts, including Bayesian decision rules, drift-diffusion models or stochastic threshold models in which the response criterion varies from trial to trial (Li et al., 2024; Ratcliff et al., 2016). These extensions may improve the modeling of reaction-time variability without changing the recurrent sensory model itself.

5 Conclusion

The main take-away is that a static within-trial prior is not sufficient to explain human reaction-time variability in this deviance-detection task. A recurrent model with state carried across trial boundaries explained additional variance, indicating that recent context can shape behavior even when the explicit stimulus rule is unchanged. At the same time, the useful recurrent models occupied a constrained region of model space and did not correspond to the lowest-loss checkpoints. Behavioral fidelity therefore depended not simply on solving the task, but on stopping at a regime whose internal dynamics remained human-like. The results support recurrent state models as a practical account of cross-trial context in auditory prediction, while leaving open the biological learning mechanism and the precise decision process that converts evolving model evidence into reaction time.

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